Testing Quick Reference

One-Command Test Everything

```
python run_tests.py
```

Individual Test Suites

```
# User registration & authentication
pytest tests/test_users.py -v

# Order creation & management
pytest tests/test_orders.py -v

# Portfolio & positions
pytest tests/test_portfolio.py -v

# Security & authorization
pytest tests/test_security.py -v

# Business logic (services)
pytest tests/test_services.py -v

# End-to-end workflows
pytest tests/test_integration.py -v
```

Load Testing (Backend Must Be Running)

```
# In terminal 1: Start backend
python main.py

# In terminal 2: Run load test
python load_test.py
```

Coverage Report

```
# Generate HTML coverage report
pytest tests/ --cov=app --cov-report=html

# Open in browser (Windows)
start htmlcov/index.html
```

```
# Open in browser (Mac/Linux)
open htmlcov/index.html
```

Run Specific Tests

```
# Single test file
pytest tests/test_users.py -v

# Single test class
pytest tests/test_users.py::TestUserRegistration -v

# Single test method
pytest tests/test_users.py::TestUserRegistration::test_register_user_success -v

# Tests matching pattern
pytest tests/ -k "rate_limit" -v
```

Test Markers

```
# Security tests only
pytest tests/ -m security

# Unit tests only
pytest tests/ -m unit

# Performance tests
pytest tests/ -m performance
```

Debugging Tests

```
# Show print statements
pytest tests/test_users.py -s

# Stop on first failure
pytest tests/test_users.py -x

# Show local variables on failure
pytest tests/test_users.py -l

# Drop into debugger on failure
pytest tests/test_users.py --pdb
```

Test Statistics

```
# Show slowest tests
pytest tests/ --durations=10

# Show test names only (no output)
pytest tests/ -q

# Show test collection
pytest tests/ --collect-only
```

Dependencies Check

```
# Install test dependencies
pip install -r requirements-test.txt

# Verify installation
python -c "import pytest; import pytest_cov; print('✅ All test dependencies
installed')"
```

Expected Output

Successful Run

```
================================= test session starts
==================================
platform win32 -- Python 3.11.x, pytest-7.4.3, pluggy-1.1.1
rootdir: c:\Project\TradingSys\Ts-backend
collected 50 items

tests/test_users.py::TestUserRegistration::test_register_user_success PASSED
[  2%]
tests/test_users.py::TestUserRegistration::test_register_user_duplicate_email
PASSED [  4%]
...
tests/test_security.py::TestSecurityHeaders::test_x_content_type_options_header
PASSED [98%]

========================== 50 passed in 12.45s
===================================
```

Coverage Summary

```
Name                     Stmts   Miss  Cover
-------------------------------------------
app/services/wallet.py    45      2    96%
```

```
app/services/order.py      78      3    96%
app/routers/orders.py      62      4    93%
-----------------------------------------
TOTAL                     185      9    95%
```

Common Issues

Issue	Solution
Import errors	Run pytest from project root
Database locked	Wait a moment and retry
Rate limit test fails	Test isolation issue, check conftest
Timeout errors	Increase timeout or check backend
Token validation fails	Ensure SECRET_KEY matches

Before Deployment

- ☐ `pytest tests/ -v` - All tests passing
- ☐ `pytest tests/ --cov=app` - Coverage > 90%
- ☐ `python load_test.py` - Load test successful
- ☐ No SQL injection vulnerabilities
- ☐ No XSS vulnerabilities
- ☐ All security headers present
- ☐ Rate limiting working
- ☐ Database transactions atomic

Files

```
conftest.py              # Test fixtures & configuration
run_tests.py             # Automated test runner with reporting
load_test.py             # Load testing script
pytest.ini               # Pytest configuration
requirements-test.txt    # Test dependencies
README_TESTING.md        # Comprehensive testing guide
TESTING_QUICK_REF.md     # This file
```

Last Updated: 2024
Total Tests: 50+
Average Runtime: 2-3 minutes